



ENHANCED HPC WORKFORCE DEVELOPMENT:

Overcoming the 'Blinking Cursor' Barrier with
Open OnDemand and projectEUREKA

PRESENTED BY:
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HiL

OUTLINE

01

Getting started in HPC and the
“Blinking Cursor”

02

OnDemand and ProjectEUREKA!

03

User Testing

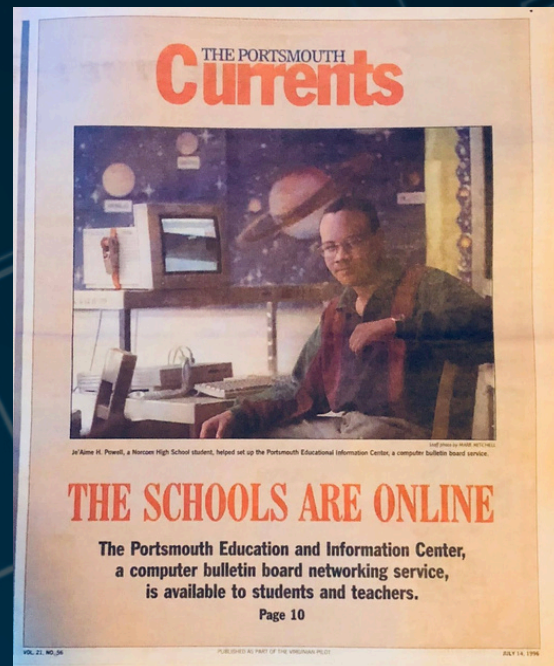
04

Outputs & Insights





Late 1900s - 2010s

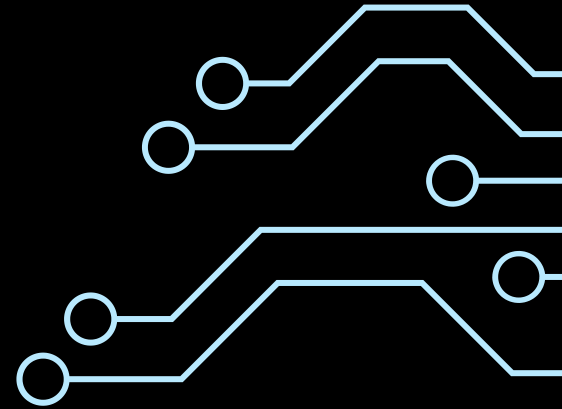


MY HPC LINUX TRAINING STORY

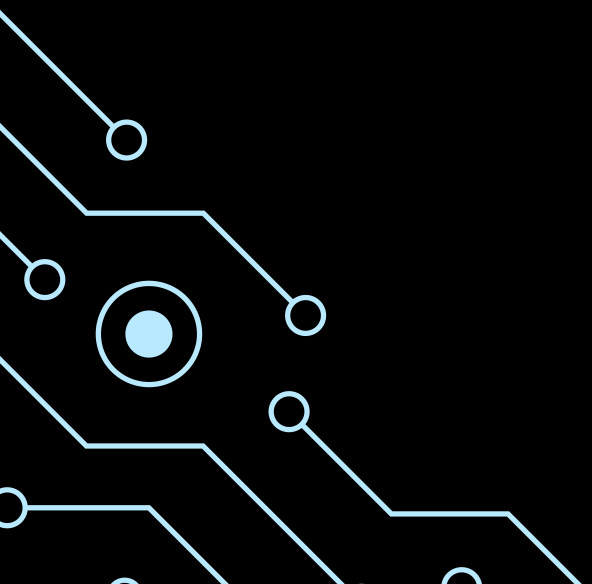
- Elizabeth City State University
- Clark Atlanta
- California State University
- ACCESS-CI/XSEDE/TeraGrid
- Hackathons
- Code@TACC
- Costa Rica National High Technology Center (CeNAT)
- Undergrad/Graduate Courses (University of Texas)
- SGX3/SGCI Coding Institutes
- UREs/REUs/NEHRIs (many)

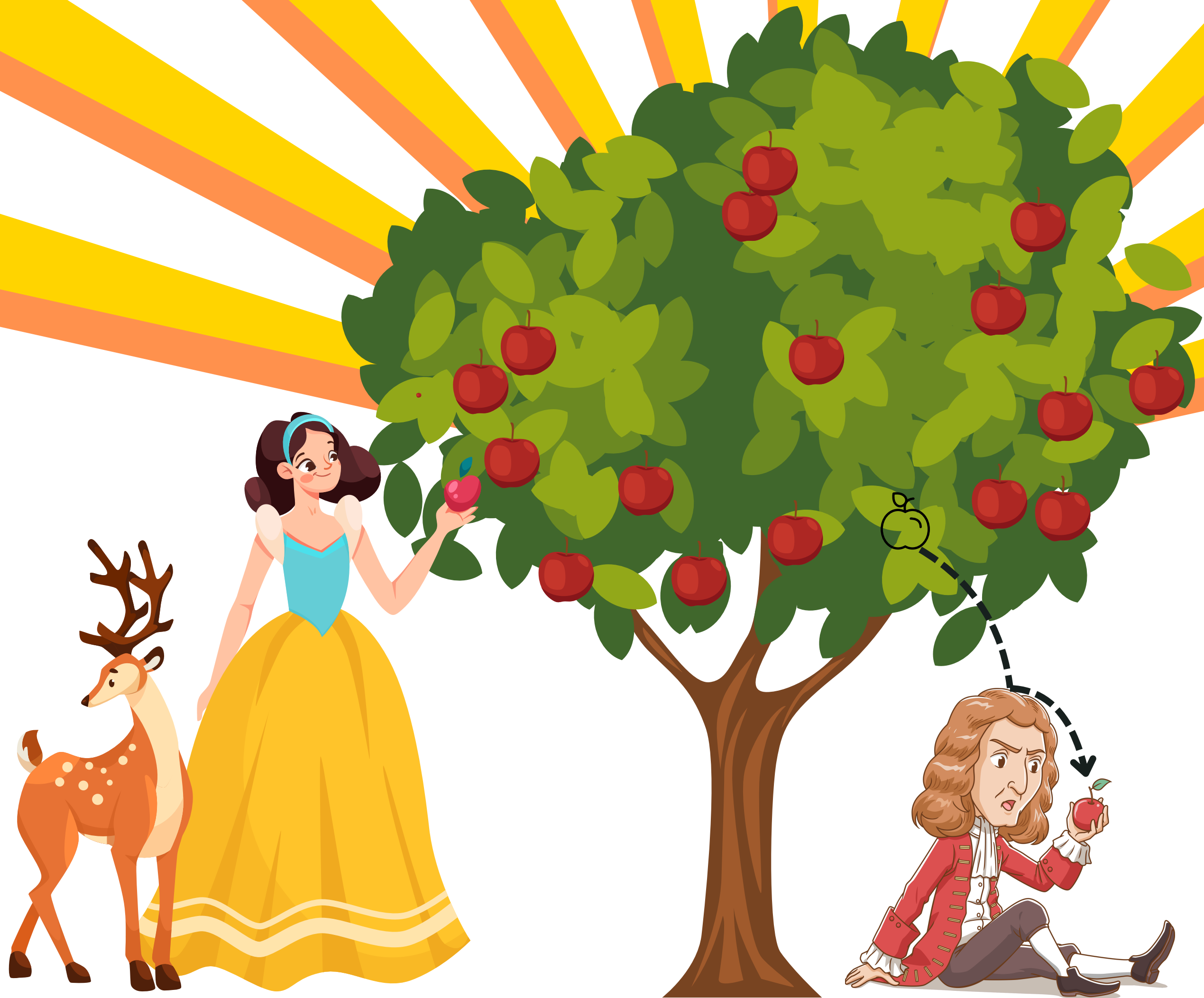
2020s





- **SSH**
- **Filesystem Tools**
- **Text Editors (VIM/EMacs/Nano)**
- **File Transfers**
- **Software Development /Installation**
- **HPC Filesystems**
- **Job Schedulers**
- **Parallel Software Development**
- **HPC Software Development**
- **Specialized Software**



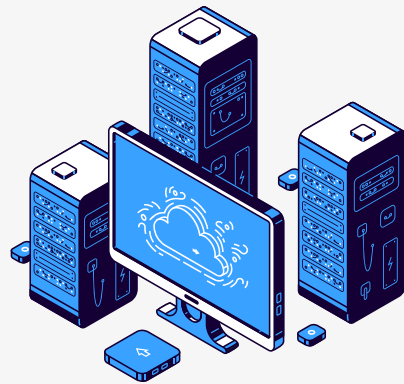




project **EUREKA!**

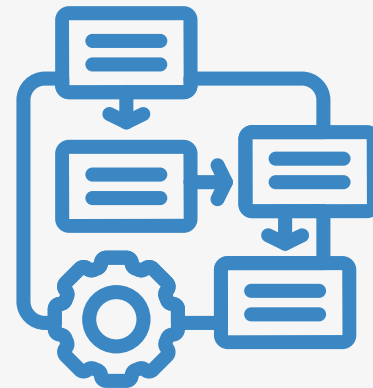


OPEN
OnDemand



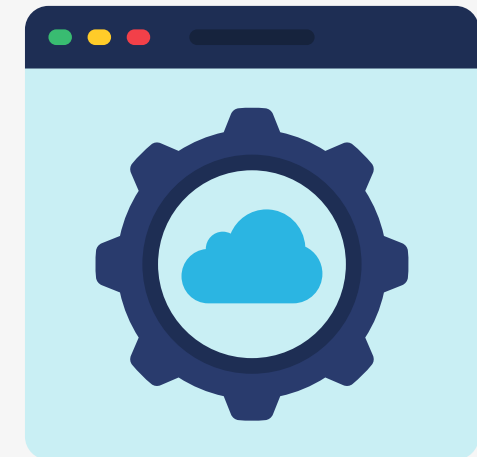
Multi-Environment
Deployment
Architecture

**Cloud/local/hybrid compute
with Kubernetes and data
staging/traditional HPC
workflows**



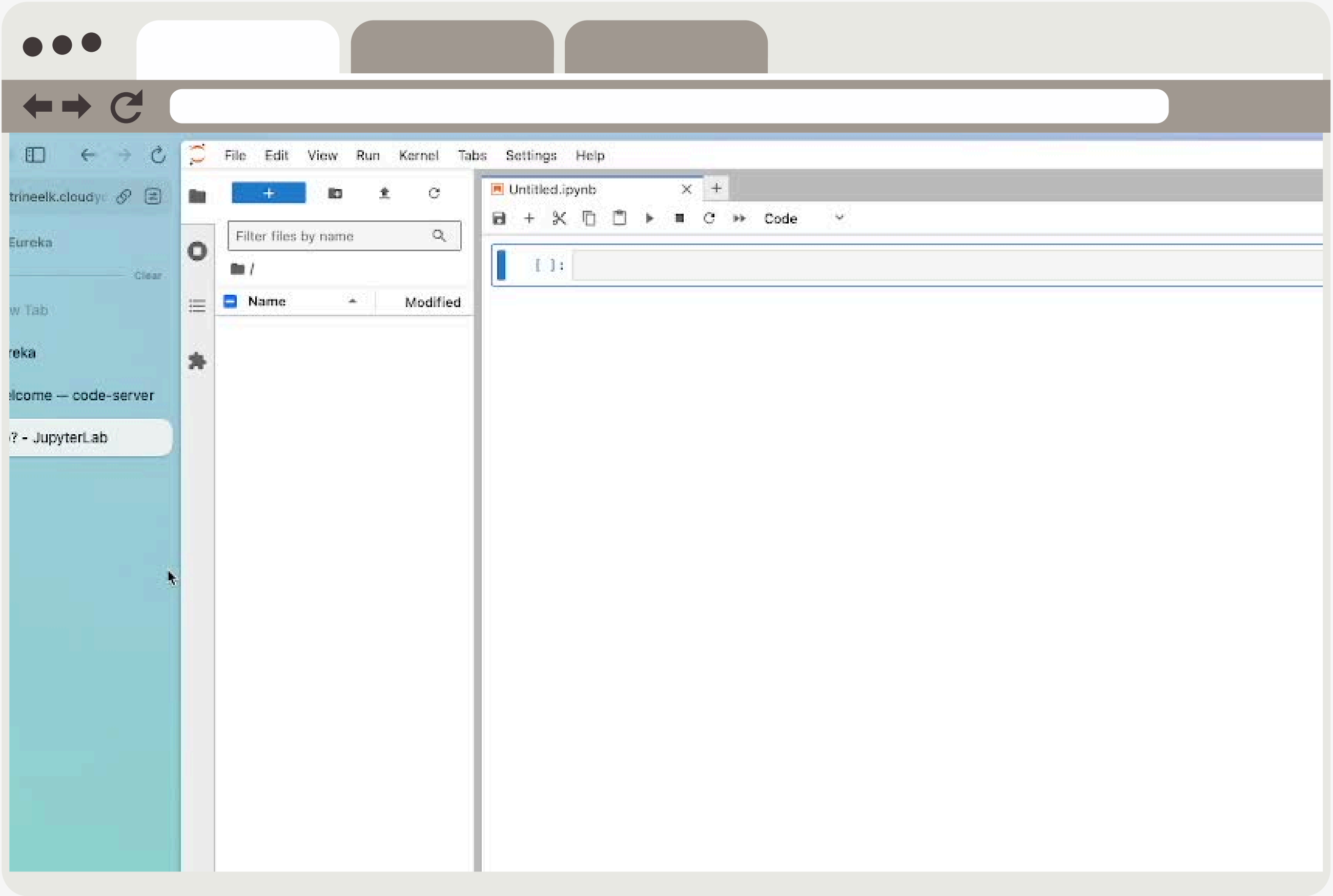
Unified AI-HPC Workflow
Orchestration and
Intelligent Data Staging

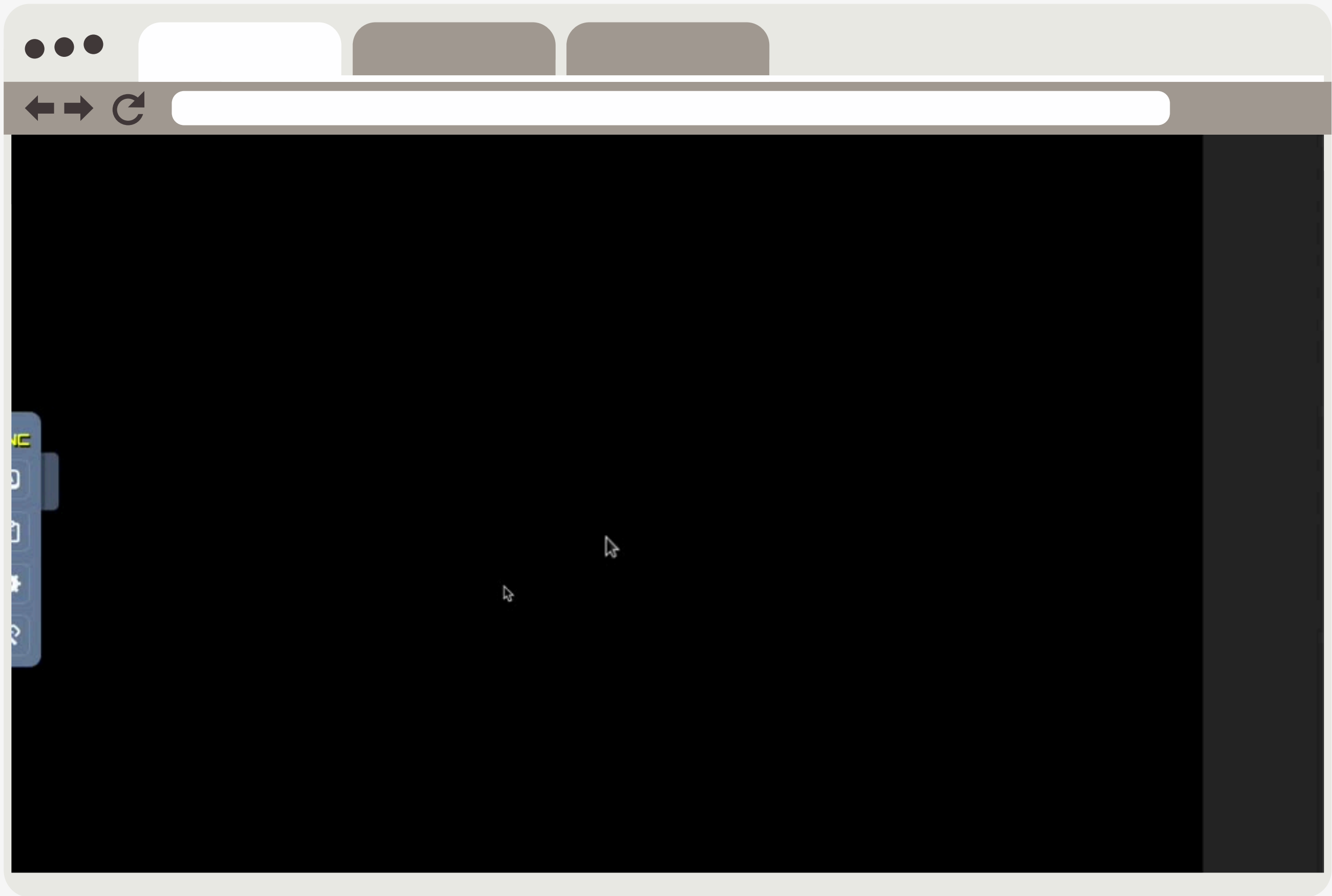
**Optimizing computational
efficiency through data-centric
job routing and automated
staging between environments**



Browser-based
Applications

**Jupyter, Desktop, R,
VSCode, PyCharm, and
more with Open
OnDemand**







User Testing through HPC/Gateways themed Hackathons

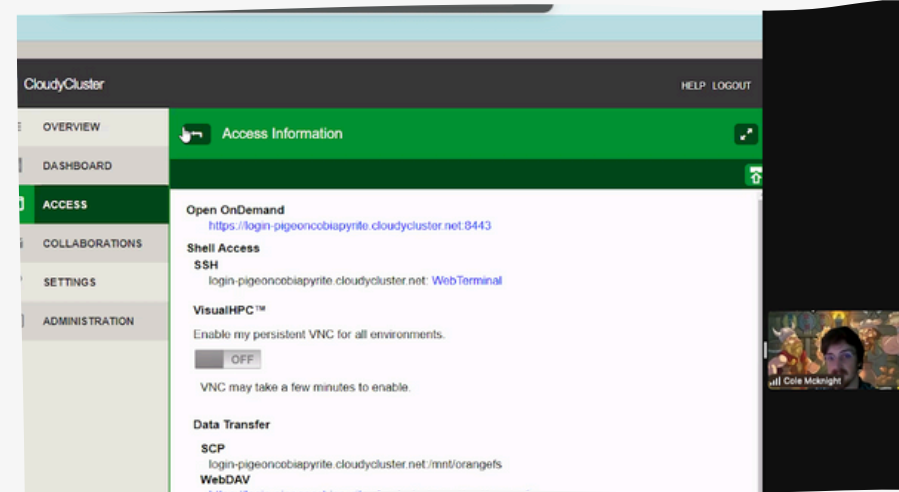
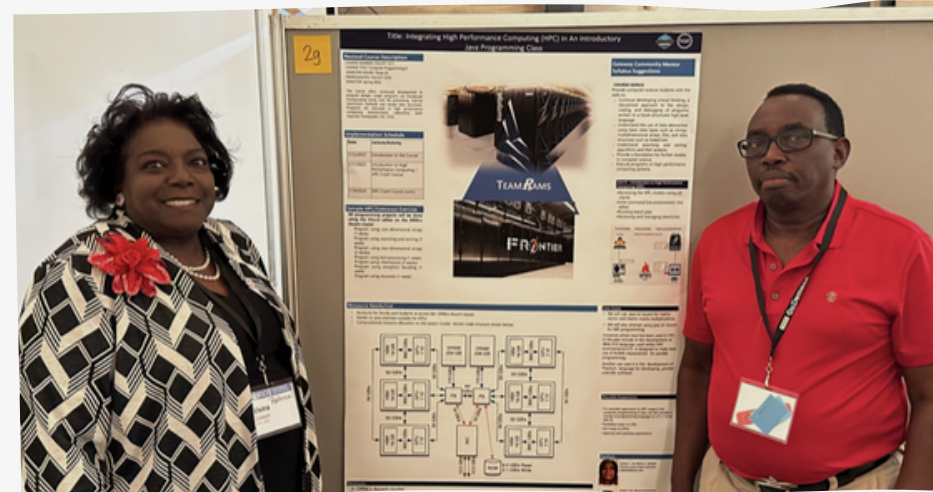


- 8 – Hackathons (3 – Faculty and 5 Student Focused)
- 250+ participants total
- Pre-Training included ~30 minute overview on projectEUREKA.

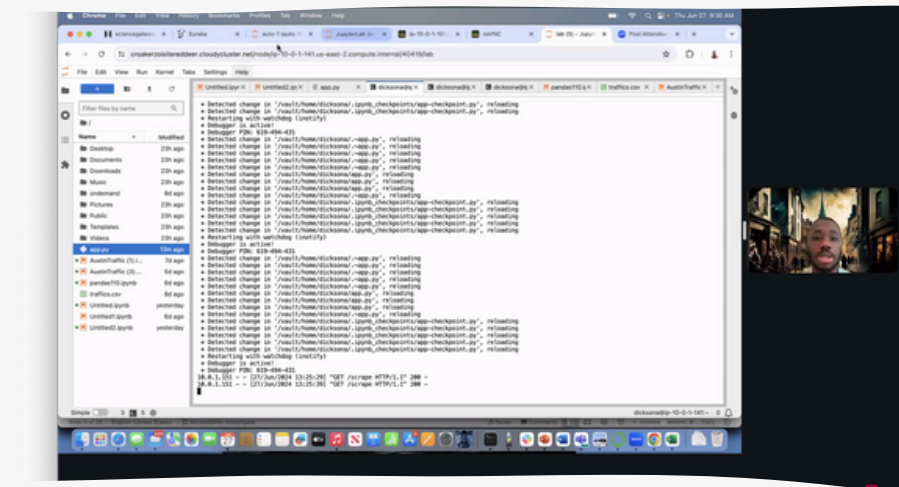
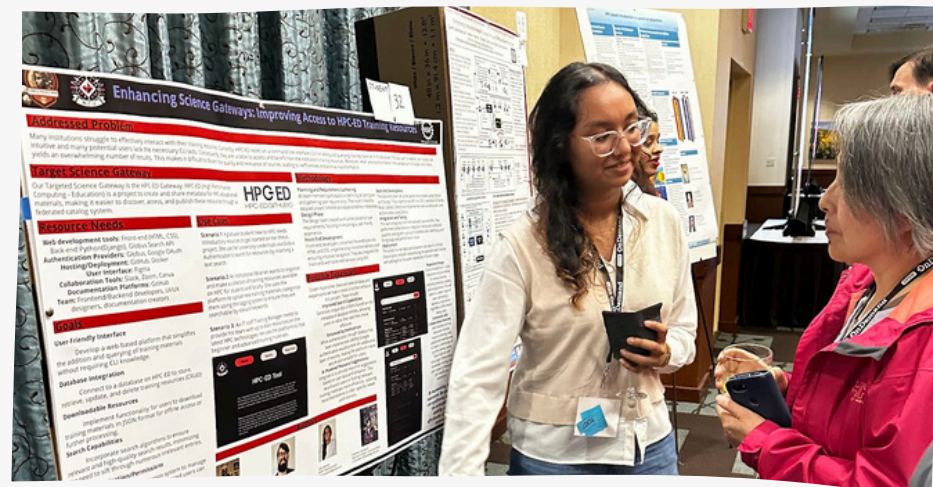




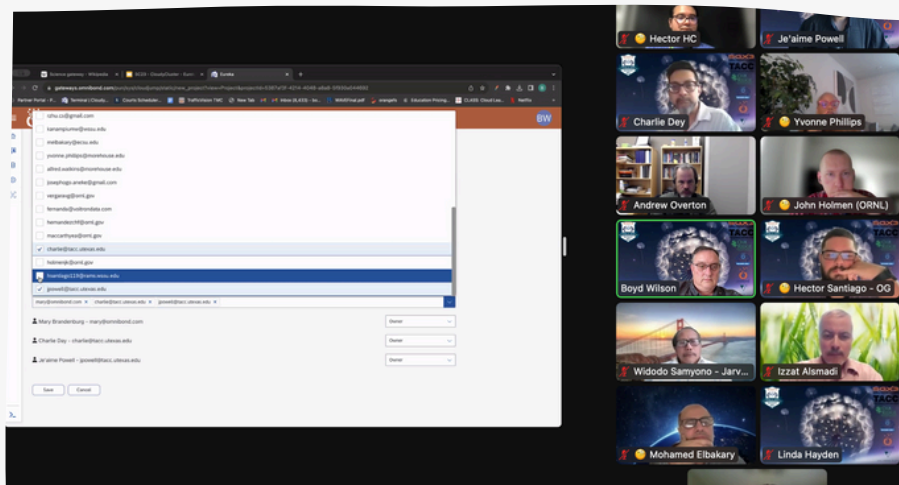
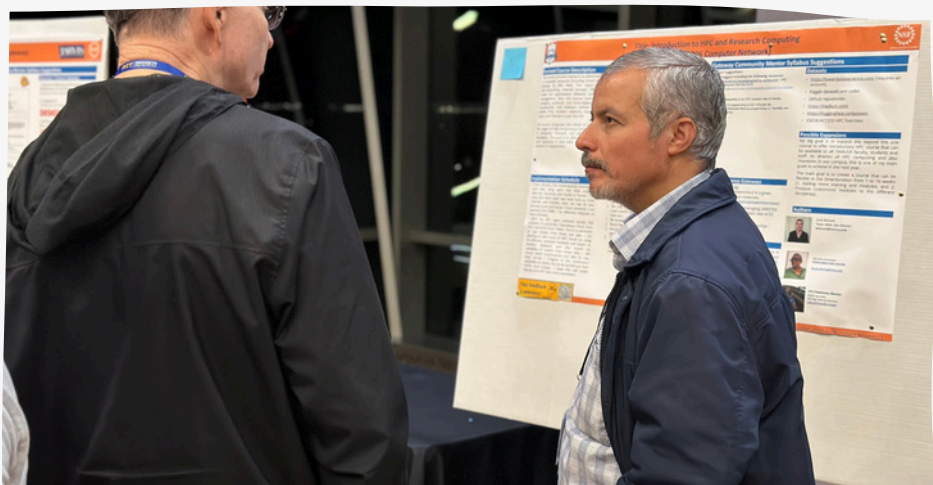
Outputs & Insights



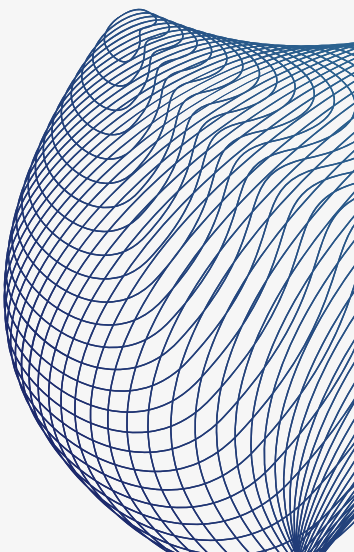
Rapid Deployment of Interactive Environments: Enabled quick launches of Jupyter notebooks, RStudio sessions, and other tools without complex setup.



Seamless Access to HPC & AI Resources: Provided direct browser-based connectivity to high-performance computing and specialized AI tools.



Enhanced Real-Time Collaboration: Facilitated resource and result sharing via a collaborative dashboard, accelerating iterative development.









projectEUREKA!

Powering your Eureka moments!



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projecteureka.ai